

Any source of heat energy is called a fuel. The term fuel is applied to a combustible substance which on burning in presence of oxygen produces a large amount of heat that can be used economically for domestic and industrial purpose.

CLASSIFICATION OF FUELS :- Can be classified on the basis of their i) occurrence and ii) Physical state

1) On the basis of occurrence - 2 Types

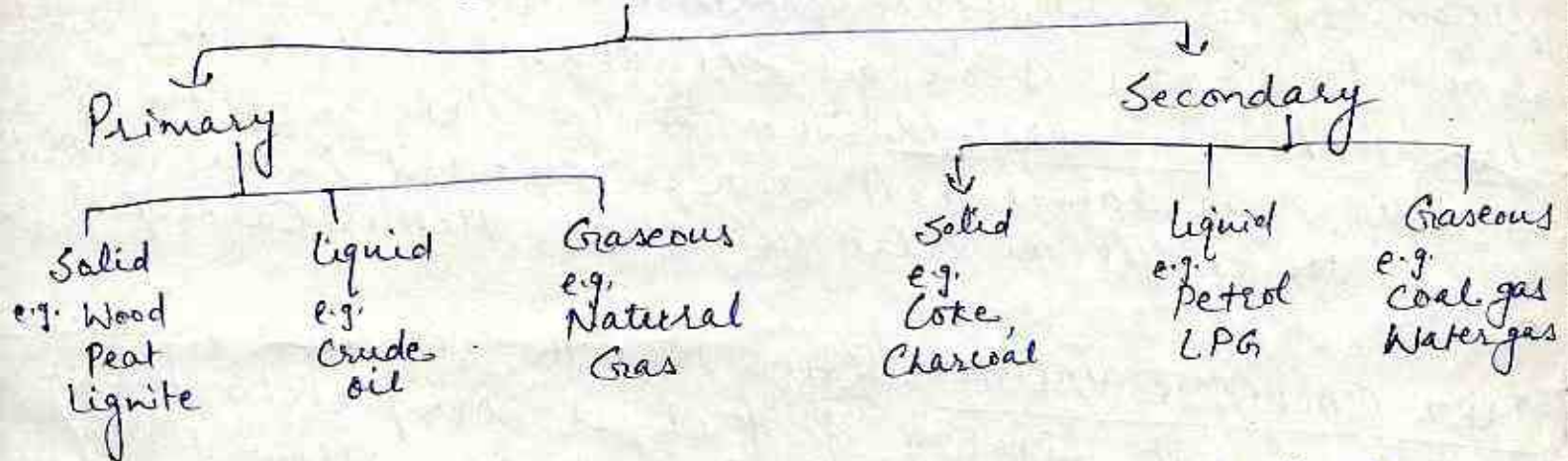
Primary fuels : Fuels which occur in nature as such are called Primary fuels. e.g. Wood, peat, coal, petroleum, Natural gas etc.

Secondary fuels : Fuels which are derived from the primary fuels by further chemical processing are called Sec. fuels. e.g. Coke, charcoal, kerosene, coal gas, Producer gas etc.

2) On the basis of physical state : 3 Types

Solid fuels
Liquid fuels
Gaseous fuels

Chemical fuels



CALORIFIC VALUE :- It tells the efficiency of fuels.

Defined as "The total quantity of heat liberated when a unit mass of a fuel is burnt completely".

UNITS OF CALORIFIC VALUE :

1. Calorie: Defined as the amount of heat required to raise the temp. of 1 gm. of water through 1°C .

$$1 \text{ Calorie} = 4.185 \text{ joules} = 4.185 \times 10^7 \text{ ergs}$$

2. KiloCalorie: Defined as the amount of heat required to raise the temp. of 1 Kg. of water through 1°C .

$$1 \text{ Kcal} = 1000 \text{ cal.}$$

3. British thermal unit: Defined as the amount of heat required to raise the temp. of 1 pound of water through 1°F .

$$1 \text{ B.T.U.} = 252 \text{ Cal.} = .252 \text{ Kilocal.}$$

4. Centigrade heat unit: Defined as amount of heat required to raise the temp. of 1 pound of water through 1°C .

$$1 \text{ Kcal.} = 3.96 \text{ BTU}$$

$$\checkmark 1 \text{ Kcal} = 2.2 \text{ C.H.U.}$$

GROSS Calorific Value: It is the amt. of heat generated when a unit quantity of fuel is completely burnt in oxygen and the products of combustion are cooled down to room temperature.

All the fuels have sufficient amount of hydrogen. The hydrogen present during combustion is converted into steam. As the products of combustion are cooled down to room temp., the steam gets condensed into water and latent heat is evolved. Thus in the determination of G.C.V., this latent heat is also included in measured heat. Hence it is also called as Higher Calorific Value.

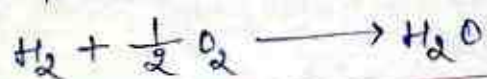
NET CALORIFIC VALUE: Defined as the net heat produced when a unit quantity of fuel is completely burnt and the products of combustion are allowed to escape.

When the fuel is burnt the water vapour, formed due to the presence of hydrogen in the fuel, are not

Condensed and escape along with hot combustion gases.². Hence a lesser amt. than Gross Calorific Value is obtained. It is also known as Lower Calorific Value.

$$\text{L.C.V.} = \text{H.C.V.} - \text{Latent heat of } \text{H}_2\text{O} \text{ vapours formed}$$

Since 1 part by wt. of hydrogen gives nine parts by wt. of water i.e.



$$\therefore \text{L.C.V.} = \text{H.C.V.} - \text{Wt. of Hydrogen} \times 9 \times \text{Latent Heat of Steam}$$

(Latent Heat of Steam = 587 Cal/gm)

Determination of Calorific Value

The Calorific Value of solids and non-volatile liquid fuels is determined by "bomb calorimeters".

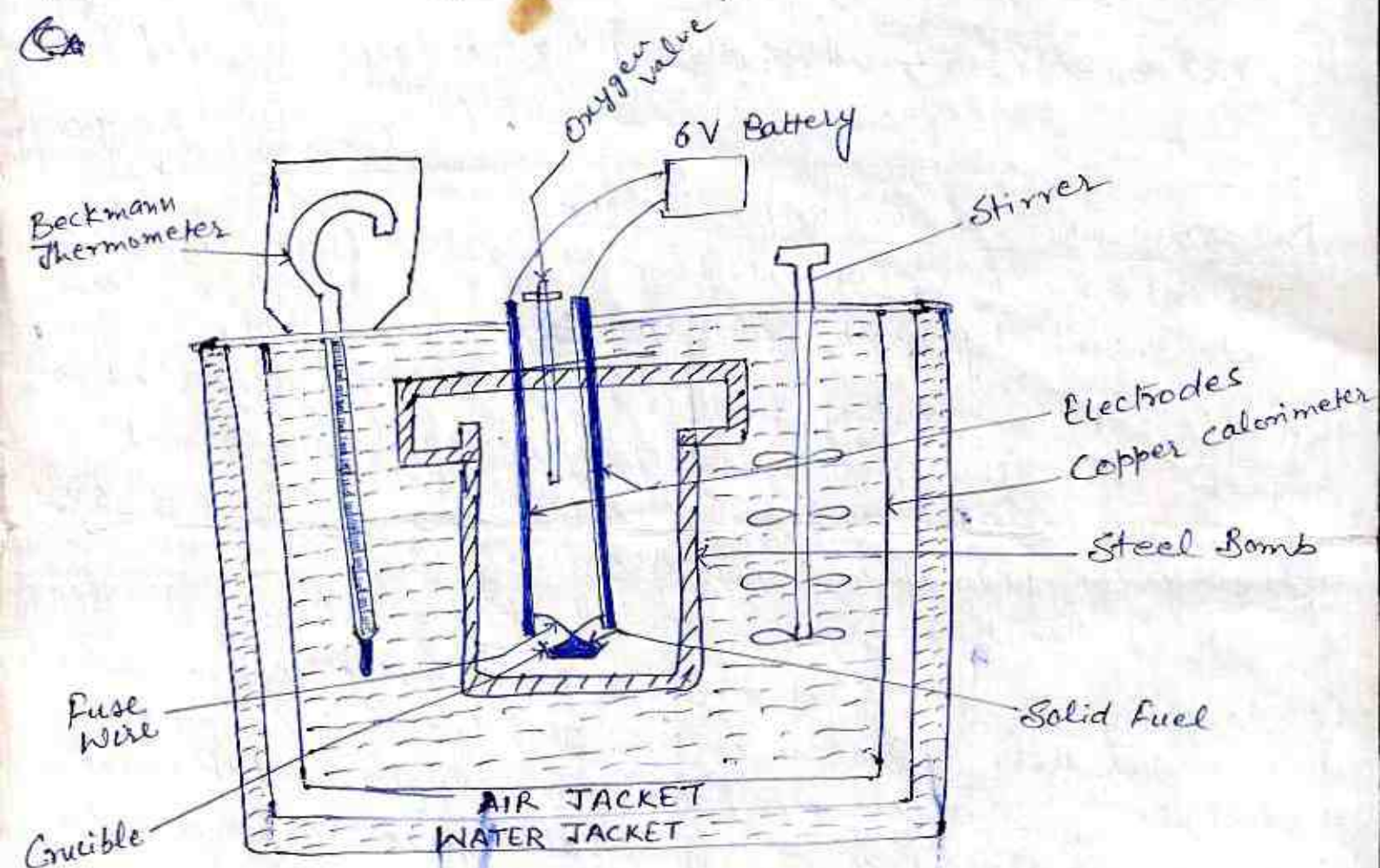
Principle: A known amt. of fuel is burnt in excess of oxygen and the heat liberated is transferred to a known amount of water. The Calorific value of the fuel is determined by applying the principle of calorimetry i.e.

$$\text{Heat Gained} = \text{Heat Lost}$$

Construction: It consists of a steel bomb which is provided with a Gas-tight screw cap or lid. The lid is provided with two holes for electrodes and one oxygen inlet valve. The bomb is placed in a Copper Calorimeter containing a known amount of H_2O . The Calorimeter is provided with a Beckmann thermometer and an electrical stirrer. The Copper Calorimeter is surrounded by the air jacket & water jacket to prevent any loss of heat. A nickel or fused silica crucible containing fuel is placed inside the ~~bomb~~ ^{bomb} attach to one of the electrodes.

WORKING:— A known amt. of given fuel (about .5 to 1g) is taken in the silica crucible. A piece of magnesium wire is tightly stretched across the electrodes, touching the fuel sample. The lid of the bomb is tightly screwed and the bomb filled with oxygen at 25 atm. pressure. The bomb is placed inside the Copper Calorimeter which contains known amt. of water.

After thorough stirring, initial temp. of the water is noted. The electrodes are then connected to a battery. The fuel in the crucible burns with the evolution of heat. The heat produced is transferred to water, which is stirred throughout the experiment. Max. temp. shown by thermometer is noted. Time taken to cool the water in the calorimeter from max. temp. to room temp. is also noted. The calorific value of



BOMB CALORIMETER

CALCULATIONS :-

Let wt. of the fuel sample taken = x g

wt. of the water in the calorimeter = W g

* Water equivalent of the Calorimeter = w g

* Water eq. of a calorimeter is defined as the no. of calories required to heat the calorimeter by 1°C . If M is the mass of Calorimeter and S is its specific heat then

$$\text{Water equivalent} = \frac{M \cdot S}{(w)}$$

Initial temp. of water = $t_1^\circ\text{C}$

Final temp. of water = $t_2^\circ\text{C}$

Gross calorific value = C cal/g

Heat gained by water = $N \times \Delta t \times \text{specific heat of } \text{H}_2\text{O}$
 $= W(t_2 - t_1) \times 1 \text{ cal.}$

Heat gained by calorimeter = $w(t_2 - t_1) \text{ cal.}$

Heat liberated by the fuel = $x \cdot C \text{ cal.}$

Heat liberated by the fuel = Heat gained by water and calorimeter
 $x \cdot C = (W + w)(t_2 - t_1) \text{ cal.}$

$$C = \frac{(W + w)(t_2 - t_1)}{x} \text{ cal/g}$$

Net Calorific Value

Let %age of hydrogen in the fuel = H

wt. of water produced from 1 gm of the fuel = $\frac{9H}{100} \text{ gm.}$

($\because$ 1 gm of hydrogen produces 9 gm of H_2O)

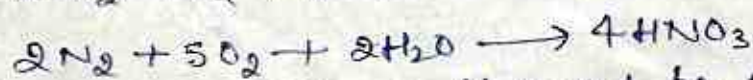
Heat liberated during condensation of steam = $0.09H \times 587 \text{ cal.}$

Net Calorific Value = G.C.V. - Latent heat of H_2O ^{vapours} formed
 $= C - 0.09H \times 587 \text{ cal/g}$

Corrections: For obtaining the accurate results, following corrections are done

a) Fuse wire correction: As Mg wire is used for ignition, heat generated by Mg wire is included in the G.C.V. Hence this amount of heat has to be subtracted from the total value.

b) Acid correction: - During combustion, Sulphur & Nitrogen present in fuel are oxidized to acids under high temp. & press.



Thus corrections should be made for the heat liberated by the formation of acids.

- for each ml of $\frac{N}{10}$ H_2SO_4 formed, 3.6 calories should be subtracted.
 - for each ml of $\frac{N}{10}$ HNO_3 formed, 1.43 cal. must be subtracted.
- (H_2SO_4 determined by pptation as $BaSO_4$ using $BaCl_2$. The corree. for 1 mg of S formed is 2.25 cal.)

(c) Cooling Correction: Heating & Cooling are simultaneous processes. As the temp rises above the room temp., the loss of heat does occur due to radiation and the highest temp recorded will be slightly less than that obtained, if no loss of heat take place. A temp. correction is therefore necessary to get the correct rise in temp.

If the time taken for the H_2O to cool down from max. temp. to room temp. = x minutes

And rate of cooling = dt/minute

Cooling correction = $x \times dt$

$\therefore$ G.C.V.

$$C = \frac{(W+w)(t_2 - t_1 + \text{Cooling corree.}) - [\text{Acid} + \text{fuse corree.}]}{\text{Mass of the fuel}}$$

Determination of Calorific Value of Gaseous & Volatile liquid Fuels: By Boy's Gas Calorimeter

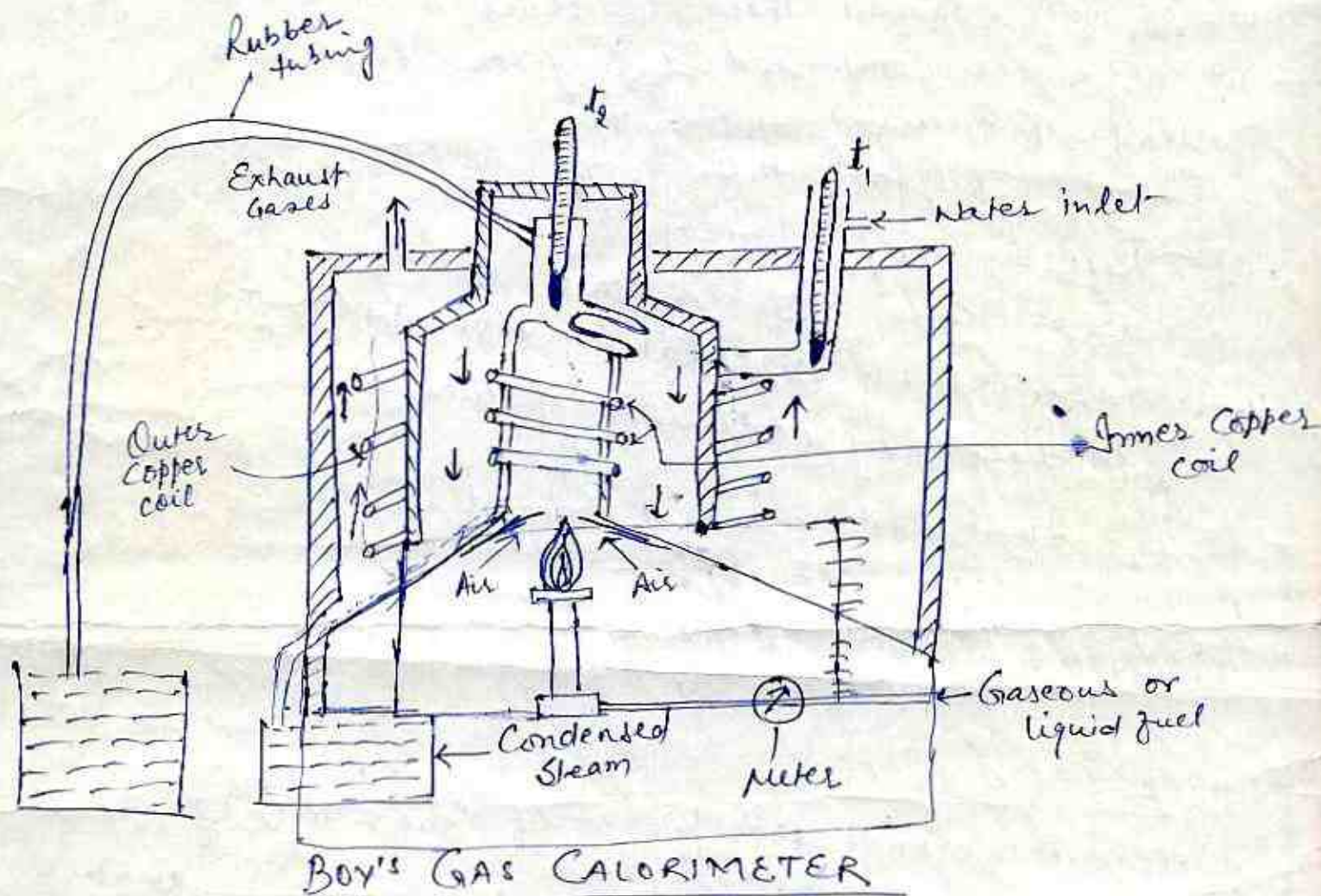
Construction: It consist of following parts:

- 1) A Gas burner ^{in which} containing a known volume of gas at a known pressure is burnt at a uniform rate. The volume of the gas is measured by a meter fitted near the gas inlet.
- 2). The burner is surr. by a chimney or combustion chamber which has a copper ~~coiling~~ ^{coil} inside as well as outside. Through this copper coil, water at a const. rate is flown. Water enters from the top of the ~~copper~~ outer coil, passes thru the outer coils, moves to the bottom of the chimney and then moves upward thru the inner coils and exit from the top.
- iii) The thermometers t_1 & t_2 are fitted to note the temp of incoming and outgoing water resp.

IV) ~~Water produced~~

A container is provided for collecting water which is formed due to condensation of steam produced during combustion.

V) whole assembly is enclosed in an insulated chamber.



WORKING: Burning of fuel and circulation of water are continued for about 15 minutes for initial warming up period. When the calorimeter gets warmed, the rate of flow is so adjusted that water leaves the apparatus nearly at atm. pressure. Heat produced by burning of fuel is transferred to water in the copper coil and the steam formed ~~inside the chimney~~ during combustion get condensed to water which is collected in a beaker. following readings are noted

- 1) Vol. of Gas burnt at a given temp. & pressure in time t .
- 2) Wt. of water passed through the coil in the same time t .

- iii) Temp. of incoming and outgoing water.
 iv) wt. of water condensed during this time.

CALCULATIONS:

Vol. of Gas burnt at STP in a time $t = V \text{ m}^3$

wt. of water passed thru the coil in time $t = W \text{ kg.}$

wt. of water condensed during this time = m

temp. of incoming $\text{H}_2\text{O} = T_1$

" " outgoing $\text{H}_2\text{O} = T_2$

Rise in temp. = $T_2 - T_1$

Gross C.V. = $C \text{ kcal/m}^3$

Heat absorbed by circulating $\text{H}_2\text{O} = W(T_2 - T_1)$

" produced by combustion of fuel = VC

Heat lost = Heat Gained

$$VC = W(T_2 - T_1)$$

$$\text{Gross C.V. } C = \frac{W(T_2 - T_1)}{V} \text{ kcal/m}^3$$

Net Calorific Value:

wt. of water condensed per m^3 of Gas = $m/V \text{ kg.}$

Latent heat of steam per m^3 of gas = $\frac{m \times 587}{V} \text{ kcal}$

$$\therefore \text{Net Calorific Value} = \left[\text{Gross C.V.} - \frac{m \times 587}{V} \right] \text{ kcal/m}^3$$

Theoretical calculations of Calorific value of a fuel

The Calorific value of a fuel can be calculated theoretically if the %ages of the constituent elements are known.

The Calorific value of the fuel is the sum of the calorific values of its constituents.

The Calorific value of C , H and S are found to be 8080, 34500, 2240 Kcal/kg resp.

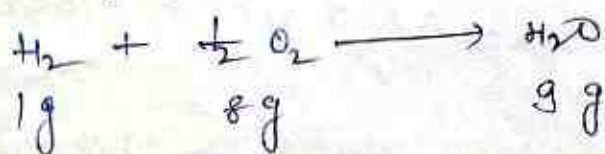
$$C = 8080, H = 34500, S = 2240$$

If Oxygen is also present, it combines with hydrogen to form H_2O . Thus Hydrogen in combined form is not available for combustion and is called fixed hydrogen.

Amt. of Hydrogen available for Combustion =

$$\text{Total mass of Hydrogen} - \text{Hydrogen combined with Oxygen}$$

Now 8 parts by wt. of Oxygen combine with 1 part by wt. of Hydrogen to form water ~~is for every~~



fixed Hydrogen = Mass of oxygen in the fuel/8

$\therefore$ Amt. of ~~oxygen~~ Hydrogen available for Combustion

$$= \text{Total mass of Hydrogen} - \frac{1}{8} \text{ mass of Oxygen in the fuel}$$

$$= H - \frac{O}{8}$$

Dulong's formula for calculating the C.V. is given as

$$\text{Gross C.V. (HCV)} = \frac{1}{100} [8080C + 34500(H - \frac{O}{8}) + 2240S] \text{ kcal/kg}$$

where C, H, O & S are %age of C, H, O & S resp.

$$\text{Net Calorific Value (LCV)} = [HCV - \frac{9H}{100} \times 587] \text{ kcal/kg}$$

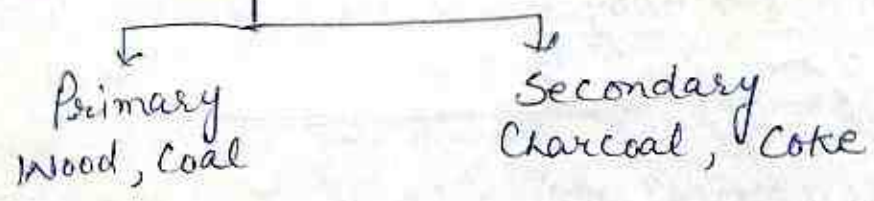
$$= [HCV - 0.09H \times 587] \text{ kcal/kg}$$

X ————— X

Characteristics of Good Fuel : following factors are taken into consideration while selecting a fuel.

- 1) Suitability : fuel selected should be most suitable for the process. e.g. Coke made out from bituminous coal is most suitable for blast furnace.
- 2) High calorific value : fuel should have high calorific value because calorific value determines the efficiency of a fuel.
- 3) Ignition temp. : It is the lowest temp. to which the fuel must be heated so that it starts burning smoothly. A Good fuel should have moderate ignition temp.
- 4) Moisture content : It should be low because high moisture content lowers the calorific value of a fuel.
- 5) Non-Combustible matter content : It forms clinkers after the combustion and reduces the heating value and also involves the cost of disposal after combustion.
- 6) Nature of the products : ^{Products formed} should not be harmful. The gaseous products evolved should not cause unpleasant odour & other harmful effects.
- 7) Cost of fuel : fuel should be available in large amount at a cheaper rate.
- 8) Smoke :- formation of smoke should be as low as possible.
- 9) Control of the process : The fuel should be such that its combustion can easily be controlled i.e. it can be easily started or stopped.

SOLID FUELS



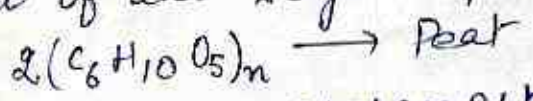
Wood: The calorific value of air-dried wood is about 3500-4500 kcal/kg. It burns with a long, non-smoky flame, leaving a small amount of ash. Presence of high oxygen content makes it a fuel of low calorific value. Still its low ignition temp, non-smoky flame and absence of ash disposal problems are the advantages. It is largely used as a domestic fuel.

Wood Charcoal: Obtained by destructive distillation of wood. It is used for producing activated carbon which finds extensive application for decolourization in Sugar Industries.

COAL:- It is a fossil fuel produced from the vegetable debris under conditions of high temp. and pressure over million of years. Chemically coal is highly complex organic matter having varying amount of water, together with nitrogen & sulphur.

The transformation of vegetable debris to coal takes place in two stages:

1) Biochemical or peat stage: During this stage, the plant materials were attacked by various microorganisms. The end product of this stage is peat



2) Chemical stage or Metamorphism: In this stage, the peat deposit buried under sedimentary deposits lose moisture and volatile components under the effect of high temp. & pressure. The peat get enriched in carbon and its oxygen content decreases. The spongy peat get transformed into hard brittle coal gradually.

CLASSIFICATION OF COAL: Coals are mainly classified on the basis of their rank, which is defined as degree of coalification from the parent material, wood. When wood is converted into coal, there

is gradual increase in the carbon content and decrease in the %age of Oxygen & Nitrogen. Depending upon the carbon content of the coal from Wood to Anthracite, coals are given ranking as in Table.

CLASSIFICATION OF COAL

Type of Coal	Carbon content %	% moisture	Calorific Value cal/gm.
Wood ↓ Peat ↓ Brown Coal ↓ Bituminous Coal ↓ Anthracite	40-50 45-60 60-75 75-90 90-95	70-90 70-90 30-50 10-20 1.5-3.5	4000-4500 4125-5280 6600-7100 6600-8800 8470-8800

The carbon content and calorific value of a coal increases with the degree of coalification, hydrogen, oxygen, nitrogen, volatile matter, moisture decreases on increase in degree of coalification.

Analysis of Coal: Coal is analysed in two ways.

1. Proximate Analysis

2. Ultimate Analysis

The results of analysis are generally reported in the following ways:

- As received basis
- Air-dried basis
- Moisture free basis (Oven dried)
- Moisture & ash free basis

Proximate Analysis: It gives information about the practical utility of coal. The data varies with the procedure adopted and hence it is called Proximate analysis. It gives information & determines the moisture, ash, volatile matter and fixed carbon of coal.

1 Moisture Content :- By origin, Coal is always associated with moisture. When a wet coal is exposed to atmosphere, the external moisture evaporates but the apparently dry coal still contains some moisture. This moisture is referred to as air-dried moisture.

Air-dried moisture is determined by heating a known amount of coal to $105-110^{\circ}\text{C}$ in an electric oven for about one hour. After one hour, it is taken out from oven and cooled in a desiccator and weighed. Loss in weight of coal is reported as moisture content on %age basis.

$$\% \text{ age of moisture} = \frac{\text{Loss in weight}}{\text{wt. of Coal taken}} \times 100$$

Significance

- 1 Moisture lowers the heating value of coal and takes away appreciable amount of the liberated heat in the form of latent heat of vaporisation. For every % of moisture present .1% of heat is lost.
 2. Excessive surface moisture may cause difficulty in handling the coal.
 3. Presence of excessive moisture ~~from~~ catches fire in the furnace.
- Hence lesser the moisture content, better is the ^{quality} ~~quantity~~ of ~~fuel~~ coal.

2. Volatile Matter : Volatile matter consist mainly of combustible gases such as H_2 , CO , CH_4 and other hydrocarbons. It consist of a complex mixture of gaseous and liquid products resulting from the

thermal decomposition of coal. It is determined by heating a known wt. of the moisture free coal sample in a covered platinum crucible at $950^{\circ} \pm 20^{\circ}\text{C}$ for 7-8 minutes. At this temp., hydrocarbons & hydrogen are escape out. The crucible is cooled, ~~first in air~~ ~~then~~ in desiccator and weighed. Loss in weight is reported as Volatile matter on %age basis.

$$\% \text{ age of Volatile matter} = \frac{\text{Loss of wt. due to removal of Volatile matter}}{\text{Wt. of Coal sample taken}} \times 100$$

Significance :-

1. High % of Volatile matter indicates that a large proportion of fuel may escape unburnt in the form of Gas or vapour.
2. The high volatile matter contains long flames, high smoke and low heating values.
3. For efficient use of fuel, the outgoing combustible gases have to be burnt by supplying secondary air. This requires a larger combustion space.
4. The high volatile coals do not cake well whereas medium volatile coals containing 20-30% of v. matter are capable of forming hard and strong coke on carbonization.
- * Coals containing less than 14% of v. matter do not cake at all and thus not suitable for manufacturing coal coke.

However in coal gas manufacture, high volatile matter content is desirable because v. matter denotes the proportion of coal which will be converted into gas and tar products by heat.

3. Ash: Coal contains mainly inorg. mineral substances which are converted into ash by chemical reactions during combustion of coal. Ash usually consist of Silica, Alumina, iron oxide and small quantities of lime, magnesia etc.

Ash content is determined by heating the residue left after the removal of volatile matter at $700 \pm 50^\circ\text{C}$ for $\frac{1}{2}$ an hour without covering. The cooled mass is weighed. From the weight of the residue, the %age of ash is calculated as

$$\% \text{ of ash} = \frac{\text{wt. of the residue left}}{\text{wt. of the coal sample}} \times 100$$

Ash can be classified as

Intrinsic Ash
Extrinsic Ash

Intrinsic Ash:- Mineral matter originally present in veg matter from which coal was formed. Consist of oxides of Na, K, Mg, Ca & Si. This is lesser in amount (about 2-3 %).

Extrinsic Ash: Mineral matter like Clay, Gypsum, dirt which gets mixed up during mining and handling of coal constitute extrinsic Ash which remains as a residue after the combustion. Extrinsic Ash may consist of anhyd. CaSO₄, CaCO₃, Fe₂O₃ etc. This is relatively high in contents.

Significance :-

- High %age of Ash is undesirable. It ~~does not~~ ^{reduces} contribute to the calorific value of coal and creates many difficulties in efficient utilisation of coal.
- In furnace, the ash restrict the passage of air and lower the rate of combustion.
- High ash leads to large heat loss and cause problem of clinkering. It has been estimated that one % rise in coal ash is equivalent to 0.3-4 % decrease in boiler efficiency.
- Its composition is of considerable importance in metallurgical operations as it affect the slag & metal composition.

④ Fixed Carbon :- Fixed Carbon Content increases gradually from lignite to Anthracite. Higher the %age of C, greater is its calorific value & better is the quality of ~~fuel~~ coal. This represents the quantity of carbon that can be burnt by a primary current of air.

%age of fixed Carbon is given by:

$$\% \text{ age of fixed C} = 100 - [\% \text{ of Moisture} + \text{Volatile Matter} + \text{Ash}]$$

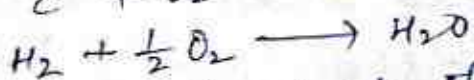
[Fixed C $\rightarrow$ This term ^{is called} fixed C because it contains other non-volatile elements also along with C]

Significance:

- ① Higher is the %age of C, greater is its calorific value.
- ② The %age of fixed C helps in designing the furnace and shape of fire box because it is the fixed C that burns in the solid state.

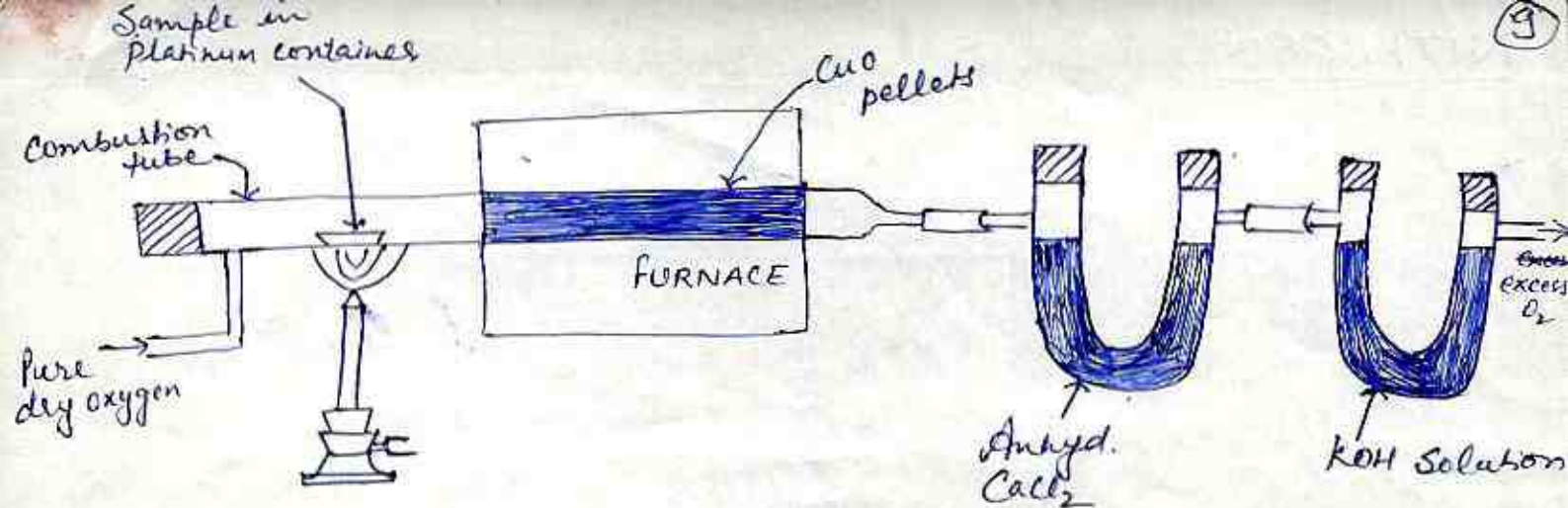
Ultimate Analysis :- It includes the estimation of C, H, S, N & oxygen. For the utilization of coal for industrial purpose, it is necessary to determine the %ages of various components.

1 Carbon and Hydrogen :- A known amount of coal is taken in a combustion tube and is burnt in excess of pure oxygen (free from moisture and CO_2). The carbon and hydrogen present in the fuel are converted into CO_2 and H_2O resp.



The gaseous products of combustion are passed thru two bulbs, one containing weighed amount of anhyd. CaCl_2 which absorbs water and the other containing weighed amount of KOH which absorbs CO_2 . The %age of carbon and hydrogen are calculated from the increase in wt. of the resp. absorption tubes.





Calculations

Let wt. of coal sample taken = x g
 Increase in wt. of CaCl_2 tube = z g
 Increase in wt. of KOH tube = y g
 44 g of CO_2 contain = 12 g of C
 y " " " " = $\frac{12}{44} \times y$ gm. of Carbon
 %age of C = $\frac{12}{44} \times \frac{y}{\text{wt. of Coal taken}} \times 100$

18 gm. of water contain = 2 g of H
 z " " " " = $\frac{2}{18} \times z$ g of H
 %age of H = $\frac{2}{18} \times \frac{z}{\text{wt. of Coal taken}} \times 100$

Significance :

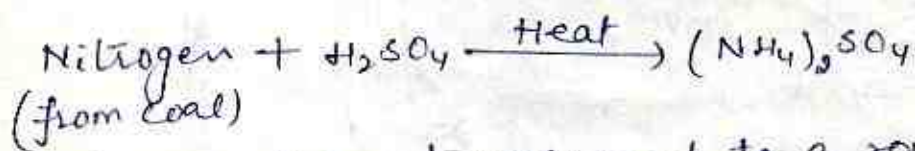
1. Calorific Value of a fuel is directly related to its Carbon content. Greater the %age of C, greater is the Calorific value of the fuel.
2. A higher %age of C also reduces the size of the combustion chamber.
3. High %age of Hydrogen also increases the C.V. of a fuel. The Hydrogen in Coal is responsible for the production of many useful materials e.g. Gaseous Hydrogen, methane and coal chemicals in Carbonization industries

NITROGEN

(10)

Nitrogen :- Nitrogen present in the coal sample can be estimated by Kjeldahl's method.

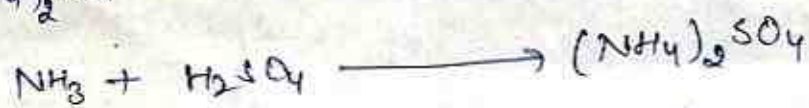
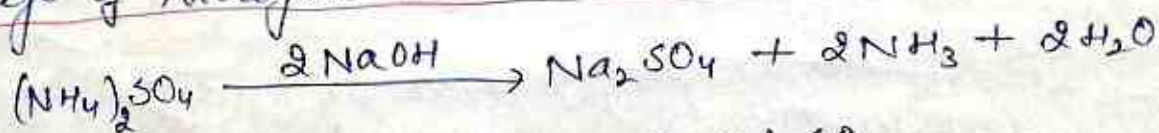
A known wt. of the given coal sample along with $K_2SO_4 + HgSO_4$ which acts as a catalyst is heated against with conc. H_2SO_4 . Nitrogen present in coal gets converted to $(NH_4)_2SO_4$, quantitatively.



The contents are then transferred to a round bottomed flask and the solution is heated with excess of NaOH.

The NH_3 gas thus liberated is absorbed in a known volume of a standard solution of acid.

The unused acid is then determined by titrating with NaOH. From the volume of acid used by NH_3 liberated, the %age of Nitrogen can be calculated.



Let wt. of the coal taken = W g

Vol. of the acid used up = V_1 mL

Normality of the acid = N_1

V_1 mL of N_1 acid = V_2 mL of N_2 NH_3

$\therefore V_2$ mL of 1N NH_3 = $\frac{V_1 \times N_1}{1}$ mL of acid

$$\begin{aligned} \text{Acid } N_1 \times V_1 &= \text{NH}_3 N_2 \times V_2 \\ N_1 \times V_1 &= 1 \times V_2 \\ \boxed{\frac{N_1 \times V_1}{1} = V_2} \end{aligned}$$

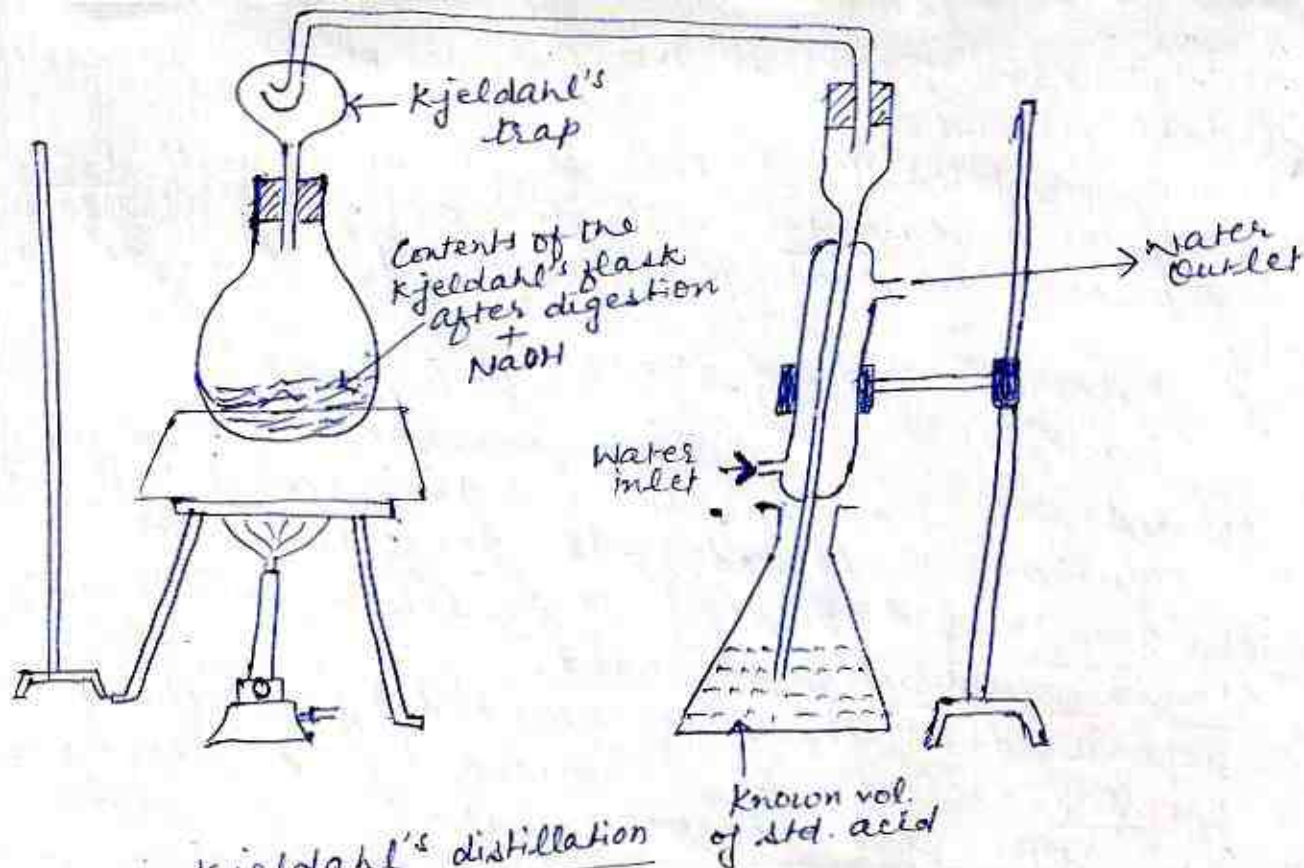
But 1000 mL of 1 normal ammonia solution $\equiv 17$ g of $NH_3 \equiv 14$ g of N

V_2 mL of 1 normal ammonia solution $\equiv \frac{14}{1000} \times N_1 \times V_1$ g of Nitrogen
($\because V_2 = \frac{N_1 \times V_1}{1}$ mL)

$$\therefore \text{Percentage of Nitrogen} = \frac{14}{1000} \times \frac{N_1 \times V_1 \times 100}{W} = \frac{1.4 \times N_1 \times V_1}{W}$$

Significance :- Nitrogen content in most of the coals is between 1 to 2 percent. It has no calorific value and hence its ~~or~~ presence is undesirable. However it is useful in carbonization industries, which recovers

Diag. $\rightarrow$



Kjeldahl's distillation

Sulphur: Sulphur present in the coal is oxidized by fuming nitric acid into sulphuric acid. The sulphuric acid is ppt. as BaSO_4 by the addition of BaCl_2 , quantitatively. The ppt. so obtained ~~from either~~ is filtered and heated to a constant weight. From the wt. of the BaSO_4 obtained the sulphur present in the coal is calculated.

Let the wt. of the coal sample = W gm.
 Mass wt. of the ppt. = W_1 gm.

$$\therefore 233 \text{ g of } \text{BaSO}_4 = 32 \text{ g of S}$$

$$W_1 \text{ " " " } = \frac{32}{233} \times W_1 \text{ g}$$

$$\% \text{ age of S} = \frac{32}{233} \times \frac{W_1}{W} \times 100$$

Significance:- Although it contribute to the C.V. of a fuel, it is undesirable constituent, because
 1. It emits corrosive SO_2 , SO_3 during combustion and highly objectionable H_2S during carbonization of coal.
 These Gases also causes atmos. pollution also.
 2. S containing coal is not suitable for the prep. of metallurgical coke as it adversely affects the properties of metal. e.g. Presence of S in the coke used in Iron Industries affects the quality & properties of Steel.

⁴ Ash :- Determined in the same way as in proximate analysis. (11) (11)
⁵ Oxygen :- O_2 is usually determined by the method of difference.

$$\% \text{ age of Oxygen} = 100 - (\% \text{ of C} + \text{H} + \text{Nitrogen} + \text{Sulphure} + \text{Ash})$$

Significance $\rightarrow$ less is the O_2 content, better is the fuel (coal).
This is because O_2 content of coal is generally associated with moisture, low calorific value, and low coking power.
Moreover, O_2 is in combined form with H_2 in the coal and thus H_2 available for combustion is lesser than the total hydrogen present.
The C.V. decreases by about 1.7% for every 1% increase in oxygen.

CARBONIZATION OF COAL (Manufacture of Coke)

It is the process of heating the coal in absence of air, to a sufficiently high temp., so that the coal undergoes decomposition and yield a residue, which is richer in carbon content than the original fuel. This residue is a lustrous, dense, strong, porous and coherent mass and is called Coke.

The process of converting coal into coke is called Carbonization.

Caking & Coking of Coals :- All types of coals cannot be converted into coke.

Caking Coal $\rightarrow$ Some coals have a tendency to soften & swell at higher temp., to form a solid coherent mass with porous struc. The residue so formed is called coke & such coals are called caking coals. e.g. Anthracite, lignite, sub-bituminous coals

Coking Coals :- If the coke is hard, porous & strong, ~~then~~ ^{then} the coal, from which it is formed, ~~then~~ ^{only} it is called coking coals. e.g. Bituminous coal can be coked.
Anthracite, Sub-bituminous, lignite coals are non-coking coals.
All caking coals are caking coals but only some of the caking coals fall under category of coking coals.

PROCESS OF CARBONIZATION :

1. First moisture and volatile gases are escape out.
2. At about 260°C - 270°C , ~~some~~ carbon, ~~some~~ H_2S and some hydrocarbons are evolved.
3. At about 350°C the decomposition of coal is accompanied by evolution of gases and elimination of tarry products take place.
4. Around 400°C , Caking coals become soft & plastic.
5. At about 700°C , Hydrogen is evolved.
6. Above 800°C , Main Gaseous products are evolved.
7. Gases evolved from the plastic mass, expand it to give foam like appearance.
8. At further high temp. the foam like mass solidifies to form a solid mass with porous struc. called Coke.

Types of Carbonization :- Properties of Coke produced depends upon

1. Type of Coal used
2. Temp. of Carbonization
3. Rate of Carbonization

Depending upon the temp., it is of 2 types :

1. Low temp. Carbonization :- When the destructive distillation of ~~the~~ coal is carried out at temp. b/w 500°C - 700°C , it is called low temp. Carbonization. LTC is carried out for the production of semi-coke, also called soft coke. ~~at cross~~

The yield of the coke is 75-80%. It contains 5-15% Volatile matter and is not sufficiently strong to be used as Metallurgical fuel.

The various products of LTC are semi-coke, low temp. tar, crude low temp. spirit & Gas. LTC plants normally use low rank ~~poor~~ coals which produce excessive smoke on burning, but the coke obtained from them is a clean fuel.

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Metallurgical coke :- The properties of coke depends upon porosity, reactivity and the amount of volatile matter retained by coke during carbonization.

- 1 porosity
- 2 reactivity
3. Amount of volatile matter ^{retained by} ~~retained~~ present in coke during carbonization.

A Good coke used in metallurgical processes should possess the following properties

- 1 Purity :- The metallurgical coke should contain lowest %ages of moisture, ash, sulphur and phosphorus.
Moisture reduces the calorific value whereas ash hinders the heating and also restrict the passage of air. High %age of ash also aids in slag formation.
S & P affect the quality of metal.

- 2 Porosity :- The coke should be porous so as to provide contact between carbon and oxygen which ensures efficient and complete combustion of the fuel in the furnace.

- 3 Strength :- The coke used in the metallurgical process should have high strength so as to withstand the wt. of the ore, flux etc. in the furnace.

- 4 Size :- Metallurgical coke should be of ^{moderate} ~~minimum~~ size - neither too big nor too small. A big size of coke does not give uniform heating. If the size is too small, it may cause choking.

- 5 Combustibility :- Coke should burn easily.

- 6 Calorific Value -> should be high.

- 7 Reactivity :- Reactivity of coke should be generally considered as its ability to react with CO_2 , steam, air & O_2 . Reactivity should not be too high.
Coke used for water gas manufacture should be reactive to steam.

Coke used in producer Gas manufacture should be reactive to CO_2



Reduction of CO_2 take place in this reaction.

- ⑧ Cost:- Coke should be cheap and easily available near the site of metallurgical plant.

Manufacture of Metallurgical Coke:-

1 By Otto Hoffmann Oven Method:- The Otto Hoffmann Oven operates on regenerative principle of heat economy i.e. Utilisation of the waste flue gases and all the valuable products are recovered from the outgoing ~~flue~~ gases.

Construction → It consist of a number of rectangular chambers made up of silica bricks. The usual dimensions are 12 to 14 m length, 4 to 5 m height and .5 m width.

① The chambers are tightly closed so that no air can enter.

② At the top of each chamber, ~~there~~ holes are provided for charging the coal.

③ Each chamber is further provided with a gas take-off ~~and coal~~ ~~in~~ ~~at each end~~ for discharging the ~~coke~~ ~~coal~~ Gas.

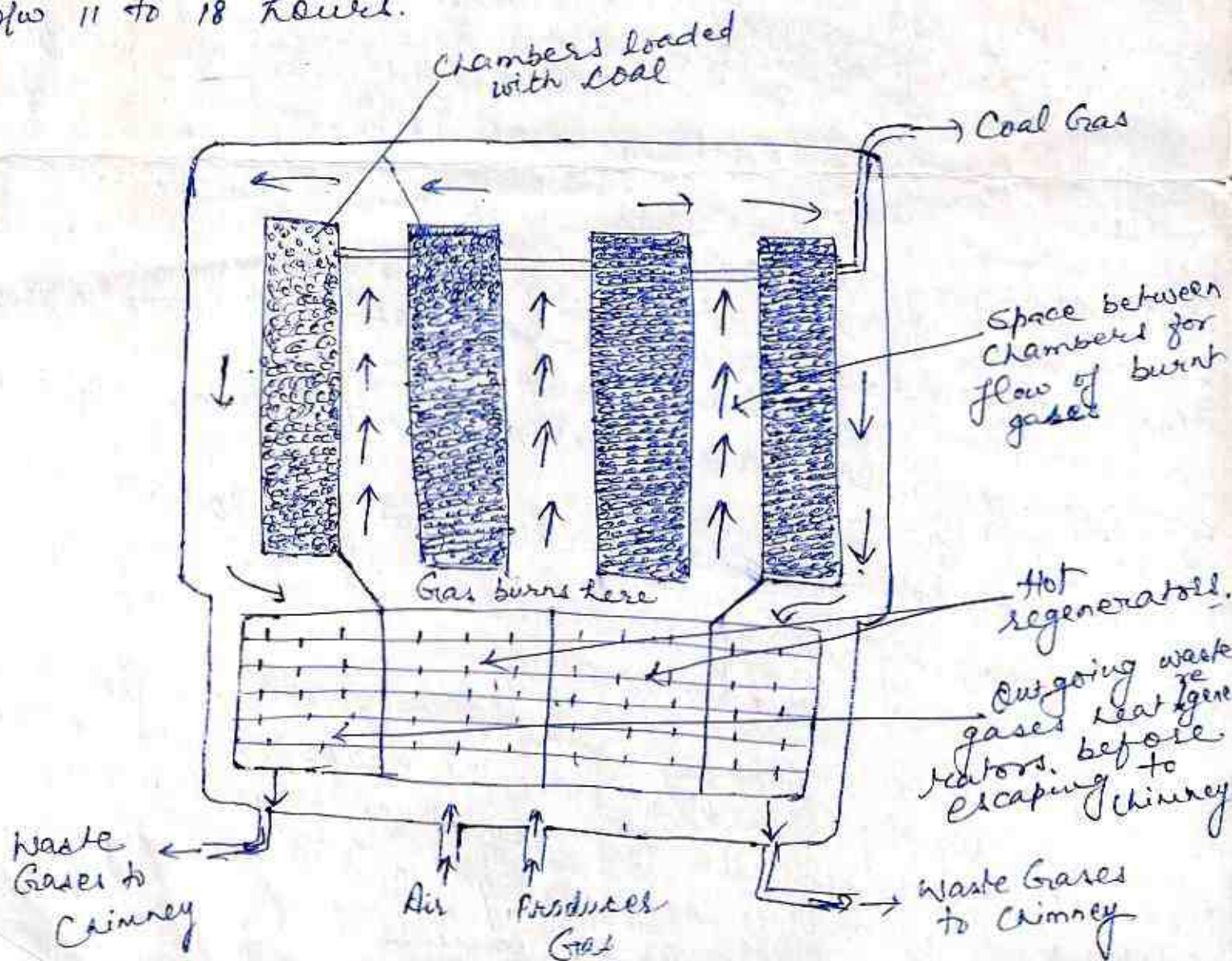
④ These chambers are erected side by side with inter space ^{for combustion} b/w them.

~~The chambers form a sort of battery of 10-100 ovens~~

⑤ One ~~big~~ single Oven or Chamber is capable of holding 16 to 24 tonnes of coal.

WORKING:- Coal is charged into the chambers and the chambers are tightly closed at both the ends. The coke ovens (like producer gas) are heated to 1200°C by burning gaseous fuel principle to achieve as economical heating as possible. The flue gases produced during combustion,

before escaping to chimney, pass on their heat to one of the two sets of checker brick-work, until ^{the temp. of} this brick work has been raised to about 1000°C . The flow of heating gases is then reversed and the inlet gases are passed through the heated checker-brick-work, which thus serves to preheat the inlet gases. The flue gases are then allowed to pass through the second set of checker ~~bricks~~ ^{brick} work to heat it. This checker ^{brick} work then serves to preheat the inlet gases, when the flow is again reversed. Thus, this cycle goes on. The heating is ~~actually~~ continued till the evolution of volatile matter ceases completely. ~~The~~ ^{The} process of Carbonization of coal takes about b/w 11 to 18 hours.



Recovery of by-products :-

The Gases coming out from the Oven is known as "Coke Oven Gas" and is mainly composed of ^{Tar,} Ammonia, ~~tar~~ naphthalene, benzene, ^{H₂S} ~~tar~~, moisture etc.

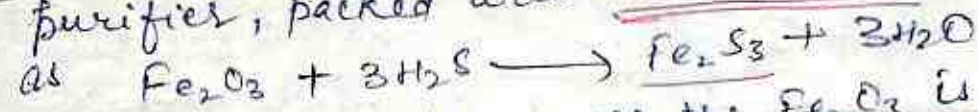
(1) Recovery of Tar :- The gas is first passed thru a tower in which liquor ammonia is sprayed. Tar and dust get collected in a tank. The tank is provided with ~~heated tank~~ heating coils to recover ~~the~~ back the ammonia. The ammonia is used again.

(2) Recovery of Ammonia :- The gases are then passed thru a tower in which water is sprayed. Here ammonia goes into solution as NH₄OH.

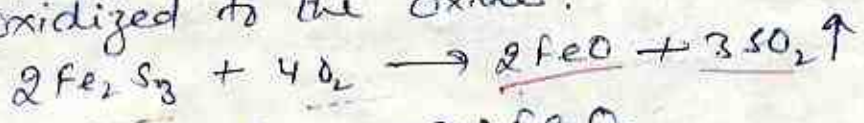
(3) Recovery of Naphthalene :- The gases are then passed thru ^{cooling tower} tower in which water at very low temp. is sprayed. Here naphthalene get condensed.

(4) Recovery of Benzene :- The gases are then ~~passed thru~~ ^{passed along with} sprayed with petroleum, when benzene ~~and~~ ^{along with} its homologues are removed & is collected at the bottom known as light oil.

(5) Recovery of H₂S :- The gases are then passed thru a purifier, packed with moist Fe₂O₃. Here H₂S is retained as

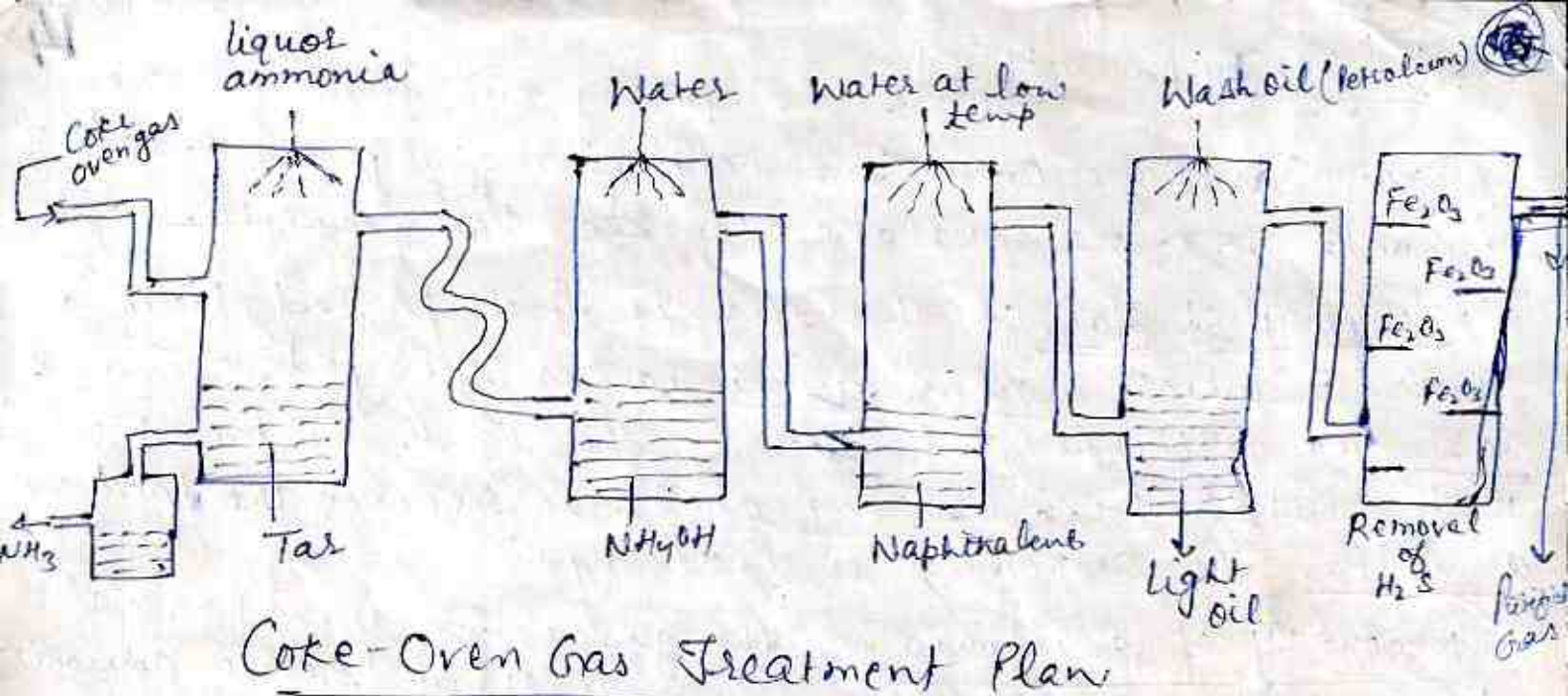


After some time when all the Fe₂O₃ is converted into Fe₂S₃ the purifier is exposed to the atmosphere when Fe₂S₃ is oxidized to the oxide.



The SO₂ obt. can be used for the manufacturing Sulphuric acid, which can be used to absorb NH₃ from the coal gas.

The purified coke oven gas is finally stored over water in holders. It is mostly used as domestic fuel in countries where it is cheap. The main components of coal gas or coke oven gas are CH₄ & H₂ along with small %age of CO₂, CO, N₂ & Oxygen.



Coke-Oven Gas Treatment Plan

Liquid fuels :- liquid fuels are used as domestic as well as Industrial ~~for~~ purposes.

The largest source of liquid fuel is petroleum. The calorific value of petroleum is about $40,000 \text{ kJ/kg}$. Though Petroleum accounts for the bulk of liq. fuels, the other sources of liquid fuels are Coal tar, Crude Benzol, Synthetic liquid fuels made from coal etc.

Petroleum :- The term Petroleum means Rock Oil. Also called Mineral oil. It is a complex mixture of Paraffinic, Olefinic and aromatic hydrocarbons with small quantities of organic compounds containing oxygen, nitrogen & sulphur.

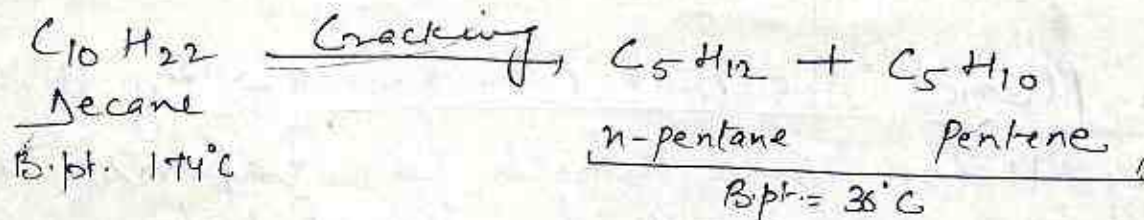
Composition → The ultimate analysis shows that the average composition of petroleum is

Element	Carbon	Hydrogen	Sulphur	Oxygen	Nitrogen
%age	80-87	11.1-15.0	0.1-3.5	0.1-0.9	0.4-0.9

The ash of the ~~at~~ crude oil is only 0.1%. or even less. Practically all metals have been found in the petroleum ash. The most widely

CRACKING :-

Cracking is the process by which heavier fractions are converted into lighter fractions by the application of heat, with or without catalyst. Cracking involves the rupture of C-C & C-H bonds. e.g.



Process of Cracking involves the following chemical changes —

- ① ~~Products~~ Higher hydrocarbons are converted into lower hydrocarbons by C-C cleavage. The products obt. on cracking have low B.pt. than the initial reactants.
- ② Formation of Branched chain HC's take place from straight chain alkanes.
- ③ Unsaturated HC's are formed from saturated hydrocarbons.
- ④ Cyclization may also take place.

Cracking ~~is~~ may be carried out by 2 methods:

1) Thermal Cracking → When cracking takes place by the application of heat and pressure, the process is called Thermal Cracking. Thermal Cracking may be carried out either in liquid phase or in vapour phase.

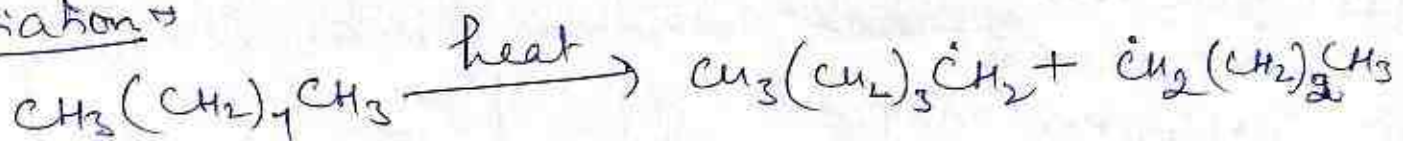
1) Liquid Phase Thermal Cracking → By this method any type of oil can be cracked. The charge is kept in the liquid form by applying high press. of the charge at a temp. of 476-530°C. The cracked products are separated in a

Fractionating column. The imp free cracked Gasoline has an octane no. of about 70. The Cracking Gases constitutes raw materials for Petrochemicals. The Cracking residues is used as a fuel.

2) Vapour Phase Thermal Cracking → By this method only those oils which can vaporise at low temp. can be cracked. The Petroleum fractions of low boiling range like Kerosene oil, are heated at a temp. of about 670-720°C. under low pressure. Petrol obt. from Vap. Phase Cracking has better antiknock properties but poor stability.

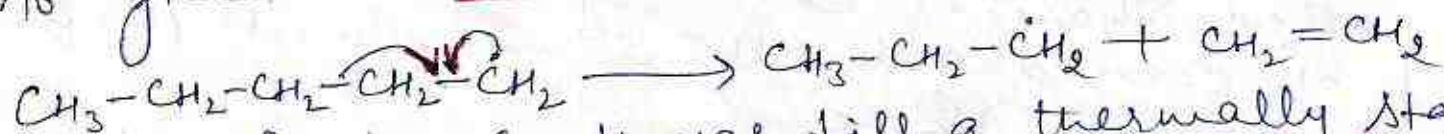
Mechanism → Thermal Cracking occurs by free radical mechanism
e.g. n-nonane

Initiation →

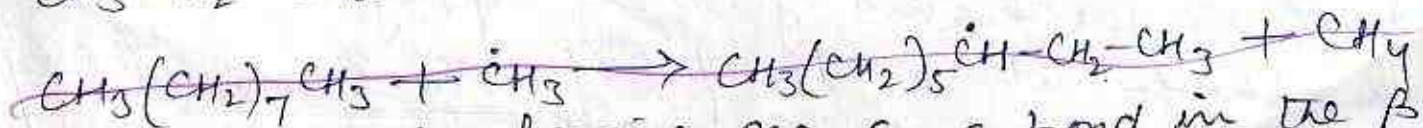
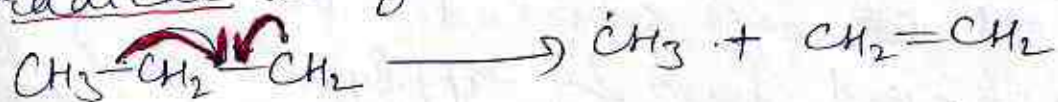


Propagation →

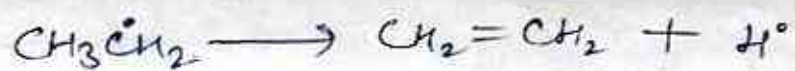
The free radicals formed are thermally unstable and undergo fission at β -position to yield a new radical and an olefin.



This β -fission continues till a thermally stable radical is formed.

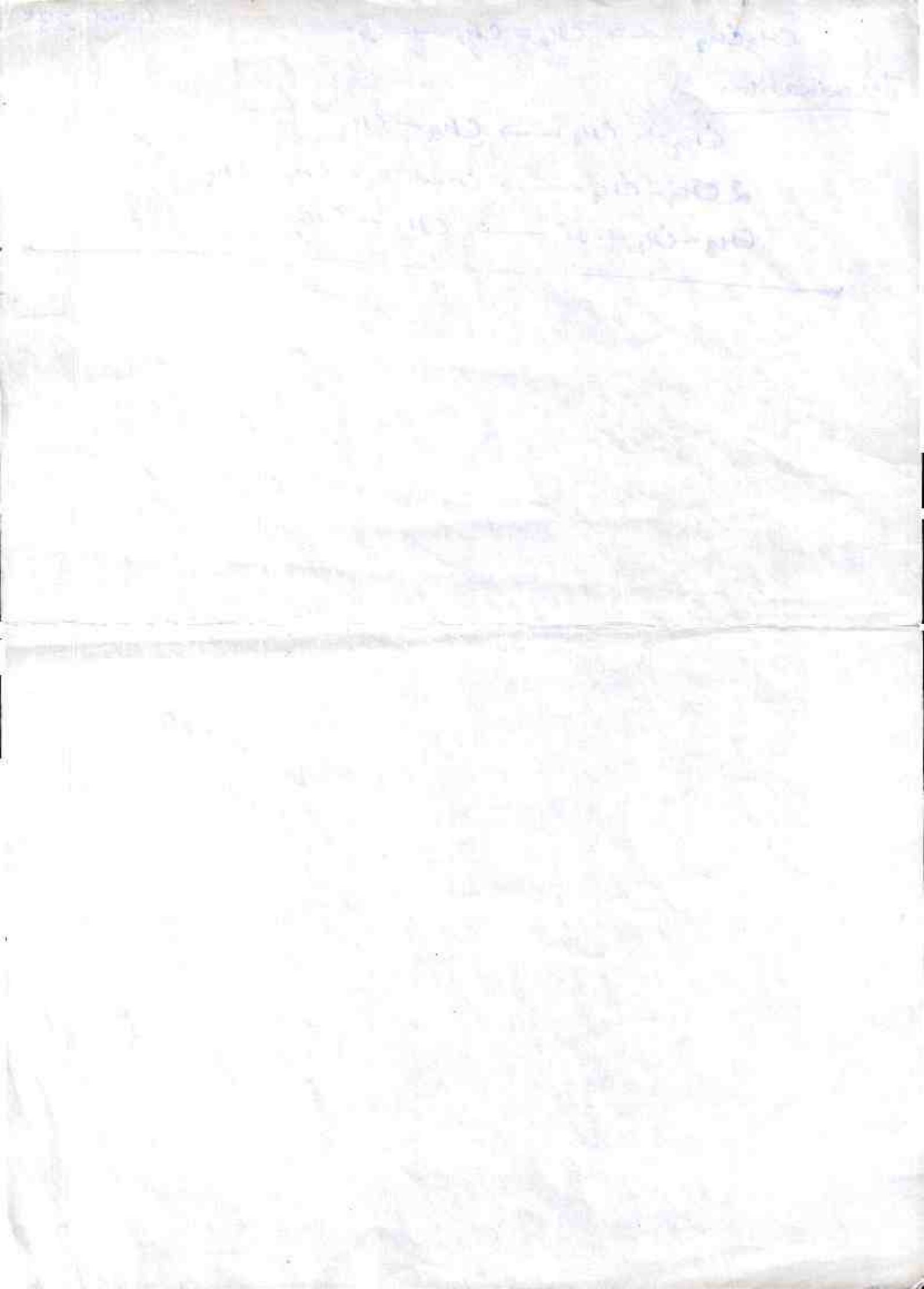


The radicals having no C-C bond in the β -position give rise to hydrogen free radical.



Termination $\rightarrow$





② Catalytic Cracking → In this method, the cracking is done in the presence of a catalyst at much lower temp. & press. The catalyst used is usually a mixture of silica & alumina (Hydrated aluminium silicate). The most recent catalyst used is Zeolite. The quality & yield of Gasoline produced by Catalytic Cracking is much improved.

Catalytic Cracking has following advantages over Thermal Cracking →

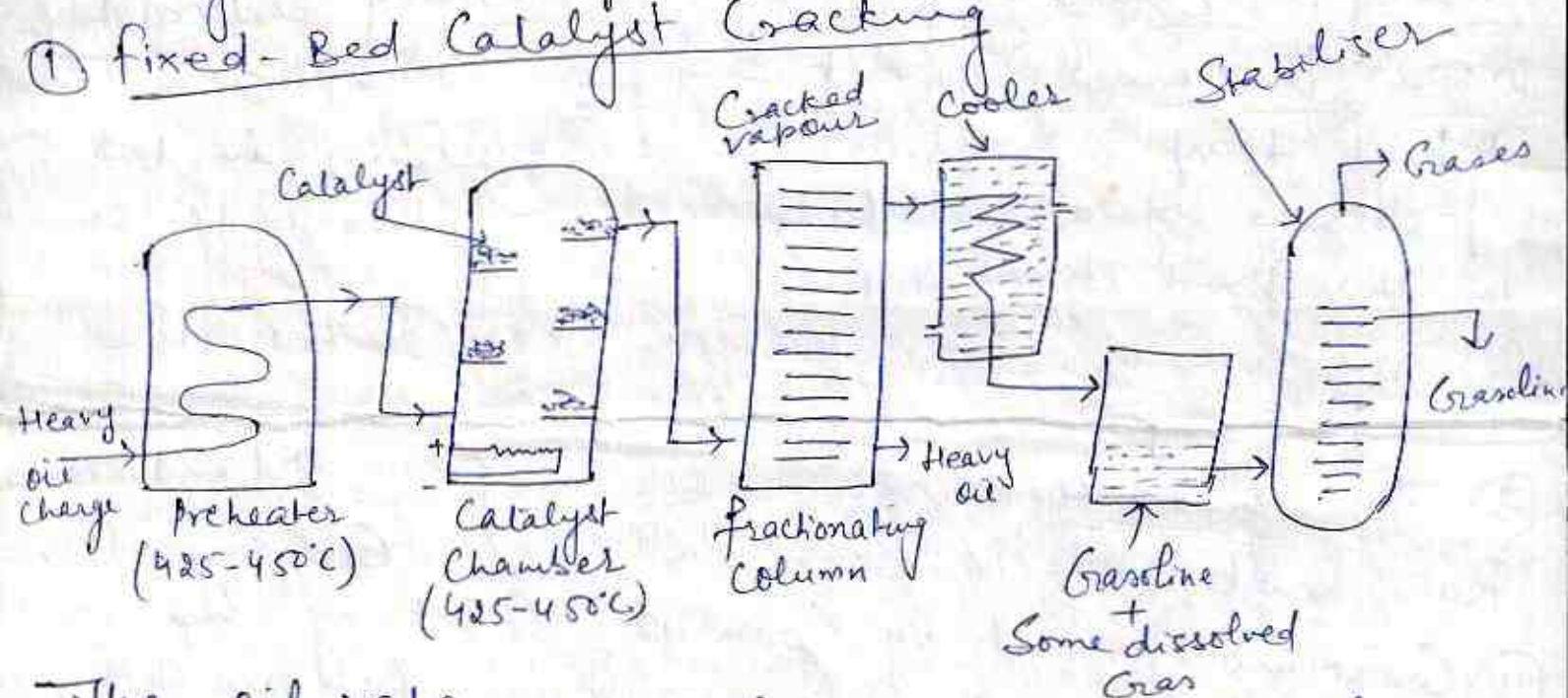
- ① High temp & press. are not required in the presence of a Catalyst & hence there is saving in the production cost.
- ② The use of ~~the~~ Catalyst accelerates the cracking rxns.
- ③ The yield of Gasoline is higher. About 70% of Raw material is converted into ~~the~~ Gasoline.
- ④ Gasoline formed by Catalytic Cracking has better antiknock properties and has an Octane no. of 80.
- ⑤ No external fuel is required for Cracking. Since the heat required is obt. by burning off the coke deposited on the Catalyst itself, during regeneration process.
- ⑥ The process can be better controlled than that of thermal process.
- ⑦ The Gasoline formed contains much less gum & gum forming compds.

⑨ The evolution of by-product gas can be minimized, by increasing the yield of desired petrol.

⑩ Catalyst are selective in action & hence cracking of only high boiling fractions take place.

Catalyst Cracking is of 2 types

① Fixed-Bed Catalyst Cracking

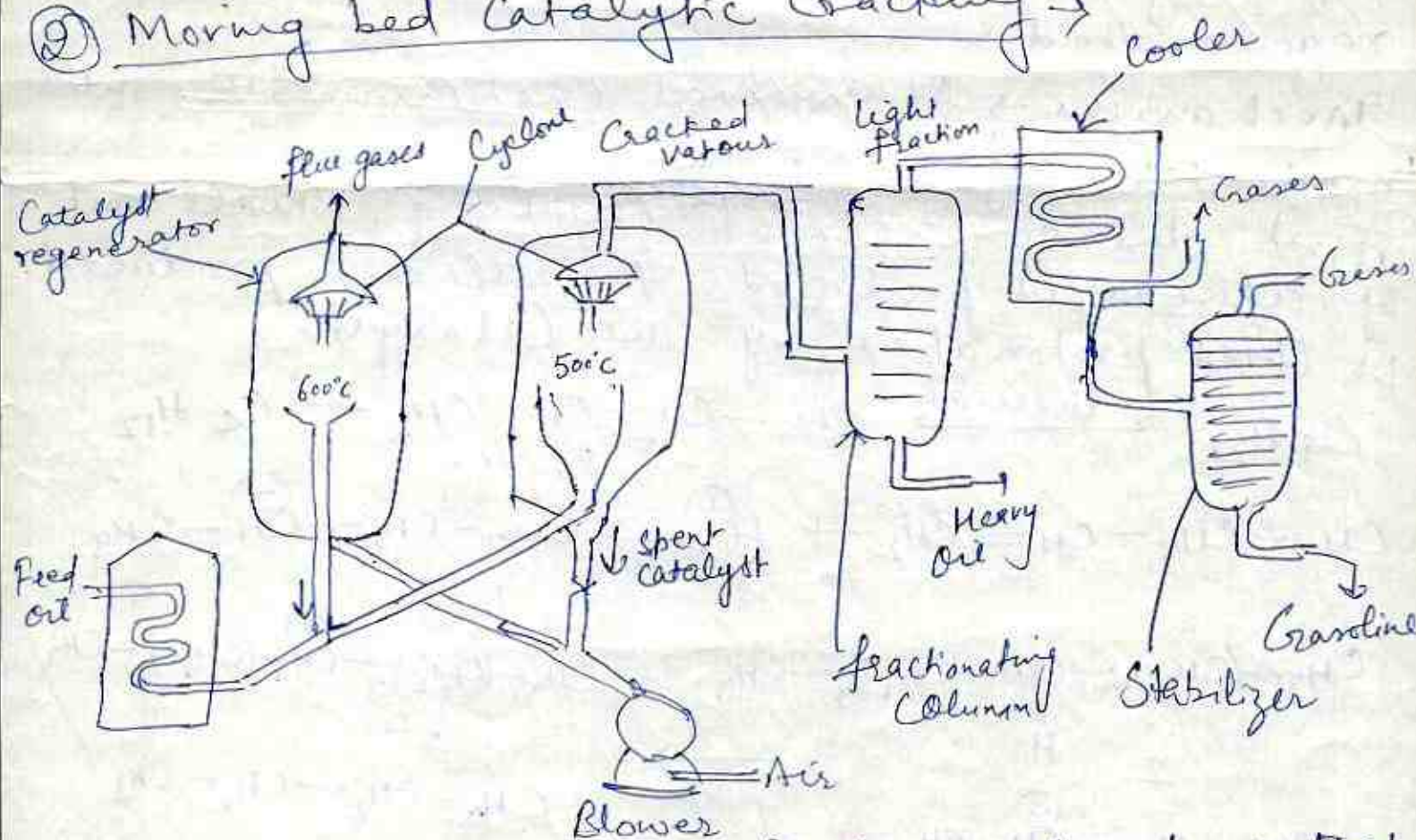


The oil-vapours are heated in a pre-heater to cracking temp. ($425-450^{\circ}\text{C}$) & then forced into a Catalytic Chamber maintained at $425-450^{\circ}\text{C}$. During their passage thru the tower, about **70%** of the charge is converted into Gasoline and about 2-4% Carbon is formed. The ~~latter~~ ^{Carbon} get absorbed on the Catalyst bed. The vapours produced are then passed thru a fractionating column where heavy oil fractions condense. The vapours are then led thru a cooler.

where some of the Gases are condensed along with gasoline and uncondensed gases ~~and~~ uncondensed gases moves on. The gasoline containing some dissolved gases is then sent to a stabilizer, where the dissolved gases are removed and pure gasoline is Obt.

The Catalyst after 8 to 10 hours, stops functioning, due to deposition of black layer of Carbon, formed during Cracking. This is reactivated by burning off the deposited Carbon.

② Moving bed Catalytic Cracking →



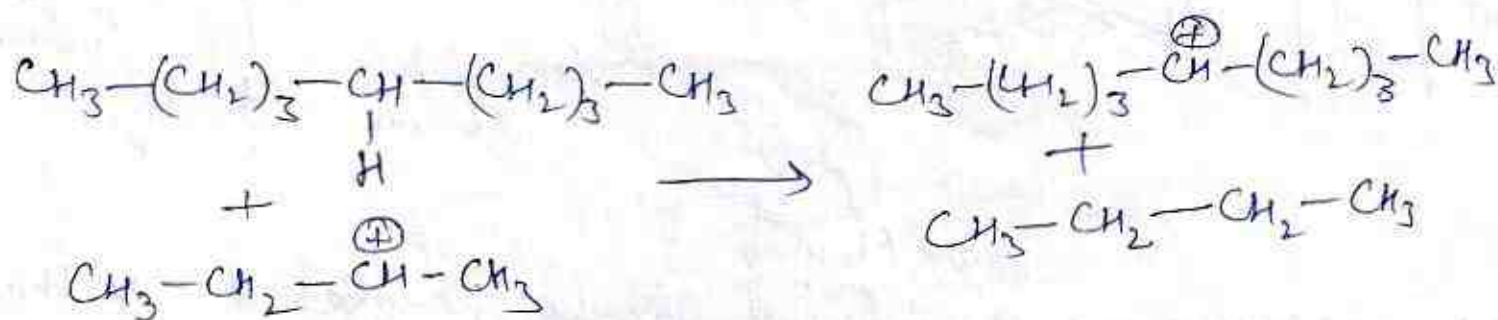
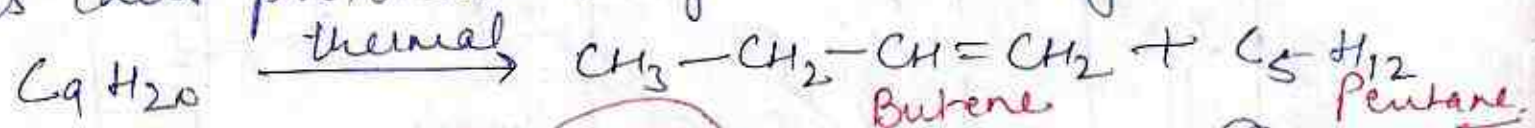
The solid catalyst is very finely powdered, so that it behaves almost as a fluid, which can be circulated in gas stream. The vapours of ~~Cracking~~ heavy oil is mixed with fluid catalyst is forced into a large reactor bed in which Cracking

of the heavier into lighter molecules occur. Near the top of the reactor, there is a centrifugal separator (cyclone) which allows only the cracked oil vapours to pass on to the fractionating column, but retain all the catalyst powder in the reactor itself. The catalyst powder gradually become heavier due to coating with carbon and settle to the bottom, from where it is forced by an air blast to regenerator (maintained at 600°C)

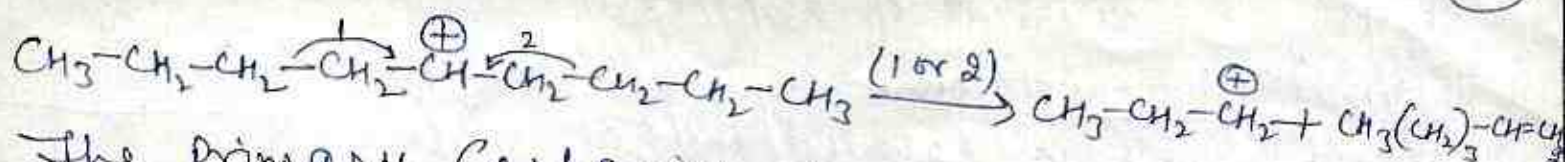
In regenerator, carbon is burnt and the catalyst is regenerated.

Mechanism $\rightarrow$ It involves Carbocation ion intermediates. e.g. n-nonane.

First due to thermal cracking, small amt. of alkene is produced. The alkene produced is then protonated by the catalyst.



The nonyl carbocation ion then undergoes β -cleavage to yield an alkene and a smaller carbocation ion.

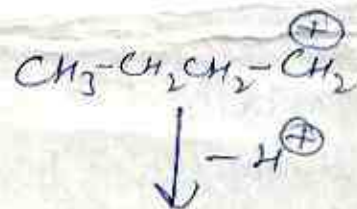
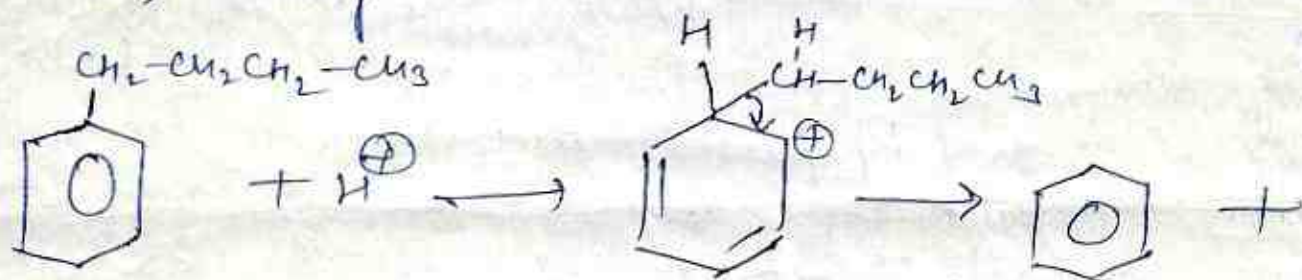


The primary Carbonium ion rearranges to the more stable secondary Carbonium ion by 1,2-hydride shift.



In aromatic hydrocarbons, the ring remains

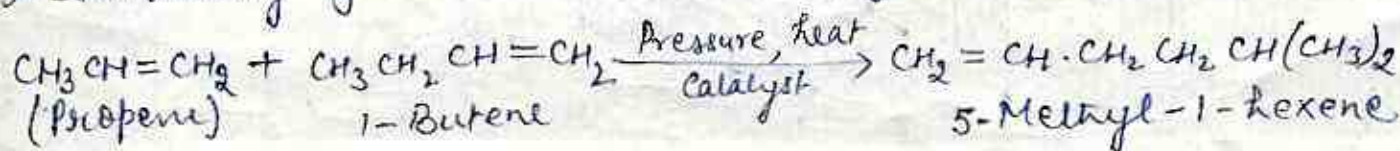
intact, only the side chains are lost



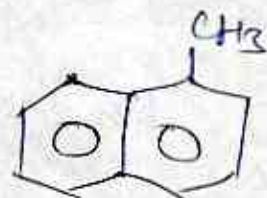
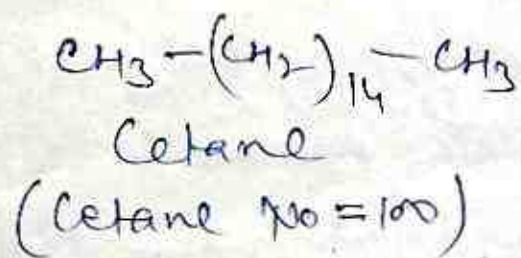
~~Not in syllabus.~~

~~Synthetic~~ Synthetic Petrol :- Petrol is synthesized by any of the following three methods :-

I. Polymerization :- The gases obt. as a by-product from cracking of heavy oils etc. contain Olefins (like ethylene, propene and butene) ^{or unsaturated hydrocarbons} and alkanes (such as methane, ethane, propane and butane). When this gaseous mixture is subjected to high pressure and high temp., with or without the presence of catalyst, it polymerises to form higher hydrocarbons resembling gasoline, called Polymer Gasoline.



Cetane No. → Hydrocarbon Cetane (n-Hexadecane) has a very short ignition delay as compared to any diesel fuel & is assigned an arbitrary value of 100. whereas α -methyl naphthalene has a longer ignition delay & hence is assigned a value of 0.



α -Methyl Naphthalene
(Cetane No = 0)

Cetane No of a ~~fuel~~ Diesel oil is defined as the % of cetane in a mixture of cetane & α -methyl naphthalene which has the same ignition characteristics as that of the fuel - under test, under same set of conditions. e.g. Diesel fuel will be assigned a cetane No. 45 if it has the same ignition characteristics as that of a mixture of 45% cetane & 55% α -methyl naphthalene.

Ignition quality order among the constituents of diesel engine fuel in order of decreasing cetane No is as follows

n -alkane $>$ naphthalene $>$ alkene $>$ branched chain alkane $>$ aromatics

In Petrol Engines $\rightarrow$ Pre-ignition by Compression leads to knocking & hence fuel with higher ignition temp. are required.

In Diesel Engines $\rightarrow$ Ignition delay is the cause of knocking and the fuels with low ignition temp. are required.

Thus a good petrol engine fuel is a bad fuel for diesel engine & vice-versa.

x-----x